
layout: cv title: Tyler Wright's Resume —

Tyler Wright

Computer Science Senior | AI Tools, Automation & Business Technology Systems

tylerwright200012@gmail.com | (916) 247-1251 | LinkedIn

Summary

Computer Science senior at BYU-Idaho focused on AI-assisted workflows, business automation, backend systems, and practical technology solutions for real organizations. Experienced building Python automation pipelines, OCR-based document processing tools, Google Drive API integrations, machine learning projects, and Elixir/OTP systems. Strong at translating business needs into tested technical solutions that improve workflow efficiency, usability, and operational reliability.

Education

2023 - Present **Brigham Young University - Idaho**, Rexburg, ID Bachelor of Science in Computer Science

2018 - 2022 **Los Medanos College**, Pittsburg, CA Associate of Science in Electrical Technology

Projects

Automated Purchase Order Document Routing System — Senior Project

- Designed and implemented an end-to-end Python automation system for processing scanned purchase order PDFs in a real automotive repair business
- Built a multi-stage pipeline for PDF conversion, OCR processing, text extraction, document classification, and Google Drive routing
- Used Tesseract OCR, Poppler/PDF2Image, regex, and spatial text analysis to identify purchase order numbers from image-based documents
- Developed rule-based classification logic for purchase orders, returns, no-PO documents, and manual review cases
- Integrated the Google Drive API to route documents into structured destination folders based on classification results
- Added logging, batch processing, and run-level summaries to support testing, troubleshooting, and operational reliability
- Built around real business requirements, edge cases, and user workflow needs to reduce repetitive manual file handling

Procession — Elixir/OTP RPG Simulation System

- Designed and developed a personal single-player RPG simulation project using Elixir and OTP as the authoritative simulation kernel
- Built architecture around supervised entity processes, message passing, world ticks, agent state, memory, dialogue constraints, and behavior validation
- Created a separation between inert world data, active simulation scopes, and live entity processes to support scalable world generation
- Implemented rule-based NPC dialogue systems that track relationship context, trust changes, topic pressure, and disclosure boundaries
- Designed AI-assisted content workflows where generated behavior metadata is treated as untrusted data until validated by deterministic Elixir systems
- Added SQLite-backed world storage to support durable world data while keeping live simulation state under OTP supervision
- Built tests and IEx/demo workflows to verify player-facing behavior, NPC responses, internal state changes, and simulation rules

Machine Learning Model Development & Evaluation

- Built and evaluated supervised learning and deep learning models using Python, pandas, NumPy, and scikit-learn

- Applied preprocessing, feature engineering, train/test evaluation, and model comparison across multiple datasets
- Used confusion matrices, precision, recall, F1 score, ROC-AUC, and overfitting analysis to explain model performance
- Created clear summaries of model results for decision-making and project reporting

Tetris Clone — Rust Individual Project

- Built an early-stage Tetris clone in Rust while learning the language independently
 - Implemented core game mechanics including piece movement, board state management, and collision detection
 - Designed modular components to organize game logic and improve maintainability
 - Used the project to strengthen understanding of Rust ownership, control flow, and structured game development
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Work Experience

Pocatello, ID Family Auto Care — Business Workflow Automation Developer

- Identified business workflow problems and developed technical solutions to reduce repetitive manual work
- Built and tested internal automation tools for document handling, file organization, and operational efficiency
- Worked directly with real business users to understand requirements, test edge cases, and improve system usability
- Supported process improvement efforts by combining technical development with practical shop operations knowledge
- Assisted with daily business operations, gaining firsthand experience with how technology affects customer service, productivity, and staff workload
- Helped connect technical improvements to practical business outcomes such as faster document handling, better organization, and reduced manual follow-up

Pittsburg, CA TOGO's — Team Member

- Worked in a fast-paced customer service environment requiring reliability, communication, time management, and attention to detail
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Technical Skills

Primary Languages: Python, Elixir, Erlang

Additional Languages: JavaScript, Rust, C++, Java, C#

Automation & AI Tools: OCR, Tesseract, PDF2Image, rule-based classification, AI-assisted workflows, local model evaluation

Web & Business Technology: JavaScript, API integrations, workflow automation, user-focused tools, business process improvement

Data & Machine Learning: scikit-learn, pandas, NumPy, preprocessing, feature engineering, model evaluation, confusion matrices

Backend & Systems: Elixir/OTP, supervision trees, process-based architecture, message passing, SQLite, command-line workflows

APIs & Integrations: Google Drive API, file routing automation, batch processing, JSON/JSONL workflows

Development Tools: Git, VS Code, IntelliJ IDEA, Linux, terminal-based development

Business & Workflow Skills: Requirements gathering, documentation, testing, troubleshooting, user support, process improvement

Volunteer Experience

2020 - 2022 **Missionary**, The Church of Jesus Christ of Latter-day Saints, Las Vegas, NV

- Served full-time in a role requiring daily planning, communication, service coordination, goal tracking, and weekly reporting
- Continued regular Japanese language study while serving
- Developed communication, discipline, adaptability, leadership, and problem-solving skills